# Retail Customer Behavior Analysis

End-to-end analytics project using Python, SQL, and Power BI

Executive project report for portfolio and GitHub presentation

This report turns the full customer shopping behavior workflow into a polished business narrative: data preparation in Python, exploratory and diagnostic SQL analysis, and interactive dashboard storytelling in Power BI.

| | |
|---|---|
| **Dataset** | customer_shopping_behavior.csv |
| **Tools** | Python, SQL, Power BI |
| **Core scope** | Revenue, customer segments, subscriptions, discounts, shipping, categories, age groups |
| **Deliverables** | Cleaned dataset, SQL query set, dashboard, findings, recommendations |

The dataset is analyzed as a transactional retail snapshot, so the report emphasizes cross-sectional customer behavior rather than seasonality or time-series trends.

# 1. Executive Summary

The analysis shows a stable average order value of $59.76 and an average review rating of 3.75 across 3,900 records. Clothing is the dominant revenue category, male customers contribute the majority of revenue, and the loyal segment is the largest customer group. Subscription status does not materially change average spend, which suggests that the current subscription value proposition may not be strong enough to lift basket size.

| Metric | Value | Meaning |
|---|---|---|
| Records analyzed | 3,900 | Base dataset size |
| Total revenue | $233,081 | Revenue across all records |
| Average purchase amount | $59.76 | Typical order value |
| Average review rating | 3.75 | Customer satisfaction proxy |
| Subscription split | 27% yes / 73% no | Subscription adoption |
| Loyal customers | 3,116 | Largest repeat segment |
| Top category | Clothing ($104,264) | Primary revenue driver |
| Top gender | Male ($157,890) | Main revenue contributor |

## High-value takeaways

| Insight | Value | Context |
|---|---|---|
| Male share of revenue | 67.7% | of total revenue |
| Clothing share of revenue | 44.7% | of total revenue |
| Loyal share of records | 79.9% | of analyzed rows |
| Avg order difference (Yes - No) | $-0.37 | subscription premium |
| Highest shipping revenue | Free Shipping | largest shipping-type contributor |

In business terms, the dataset points to a few clear actions: protect and scale Clothing, use loyalty-led retention for the large repeat base, and refine subscription and discount strategy to improve margin and conversion rather than simply expanding volume.

## 2. Methodology and Data Preparation

The workflow followed a standard analytics pipeline. Python was used to inspect the raw CSV and create a derived age_group column using quartile segmentation. SQL was then used for business questions that required aggregation, ranking, filtering, CASE logic, and a common table expression. Power BI was used to turn the results into an interactive dashboard with slicers for subscription status, gender, category, and shipping type.

| Stage | What was done | Why it matters |
|---|---|---|
| Python | Data loading, validation, and age quartile creation | Prepared the analysis-ready frame |
| SQL | GROUP BY, CASE, CTE, ROW_NUMBER, subqueries | Answered business questions |
| Power BI | KPI cards, category charts, slicers, interactive views | Delivered stakeholder-friendly visuals |

### Analytical design choices

The age groups were built from quartiles so the segmentation stays balanced instead of using arbitrary age breaks. Customer segments were defined from previous purchase counts, which makes the loyalty analysis easy to interpret for a retail audience. Each SQL question was chosen to map directly to a business decision rather than a purely technical metric.

## 3. Revenue and Product Performance

Category-level performance shows a clear concentration of revenue in Clothing, followed by Accessories, Footwear, and Outerwear. Transaction volume follows the same order, which indicates that Clothing is not only higher value per transaction but also the most frequently purchased category.

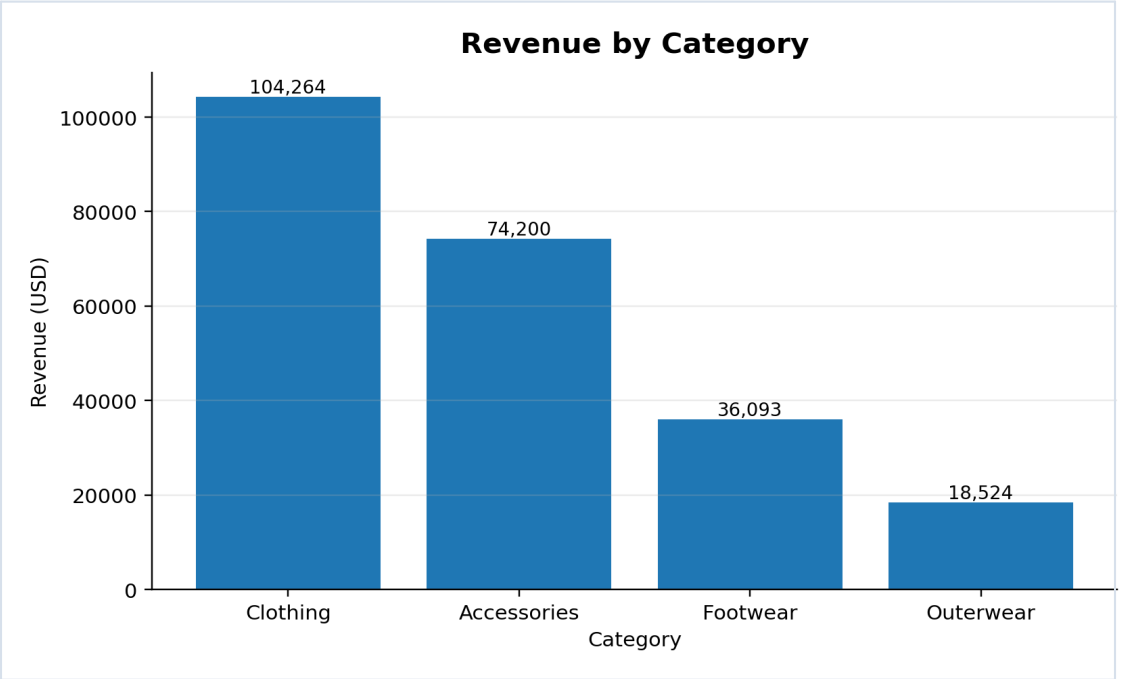


*Figure 1. Revenue by category. Clothing is the clear leader and should remain the primary merchandising focus.*

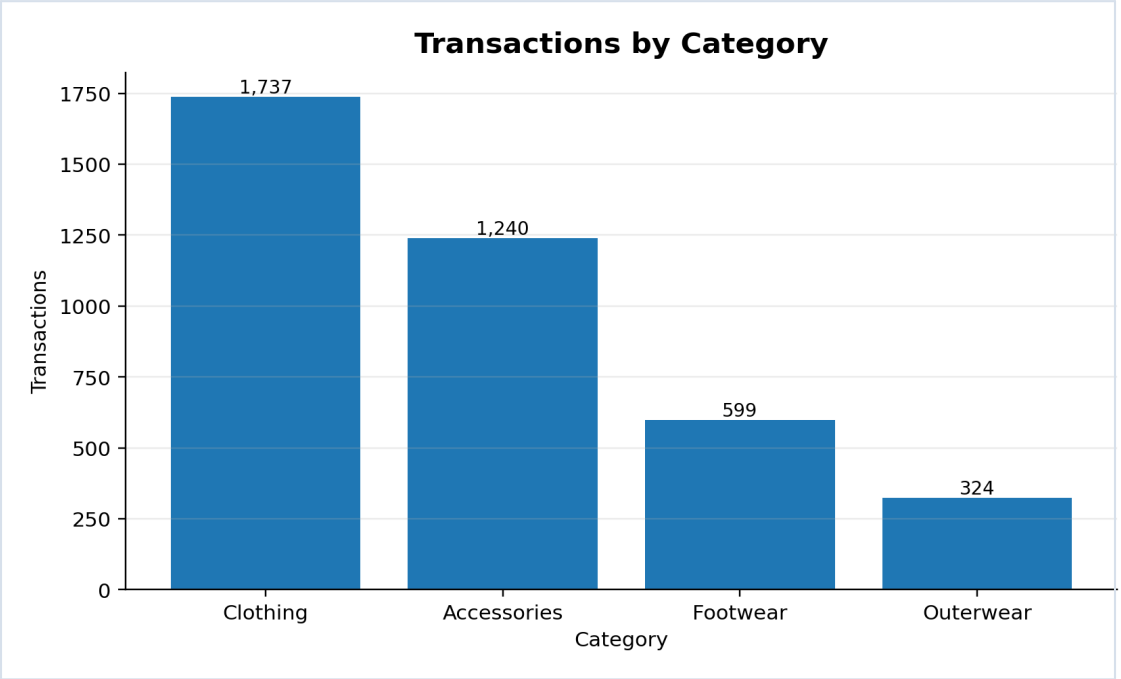


*Figure 2. Transaction count by category. The ranking mirrors revenue, reinforcing Clothing as the most important commercial segment.*

| Category | Revenue | Transactions | Revenue share |
|---|---|---|---|
| Clothing | $104,264 | 1,737 | 44.7% |
| Accessories | $74,200 | 1,240 | 31.8% |
| Footwear | $36,093 | 599 | 15.5% |
| Outerwear | $18,524 | 324 | 7.9% |

Top product revenue is concentrated in a small set of items, with Blouse, Shirt, Dress, Pants, and Jewelry leading the list. This supports a cross-sell strategy around high-volume apparel items and suggests that premium add-ons can be paired with the most demanded products.

| Top product | Orders | Avg rating | Revenue |
|---|---|---|---|
| Blouse | 171 | 60.88 | $10,410 |
| Shirt | 169 | 61.14 | $10,332 |
| Dress | 166 | 62.17 | $10,320 |
| Pants | 171 | 59.01 | $10,090 |
| Jewelry | 171 | 58.54 | $10,010 |

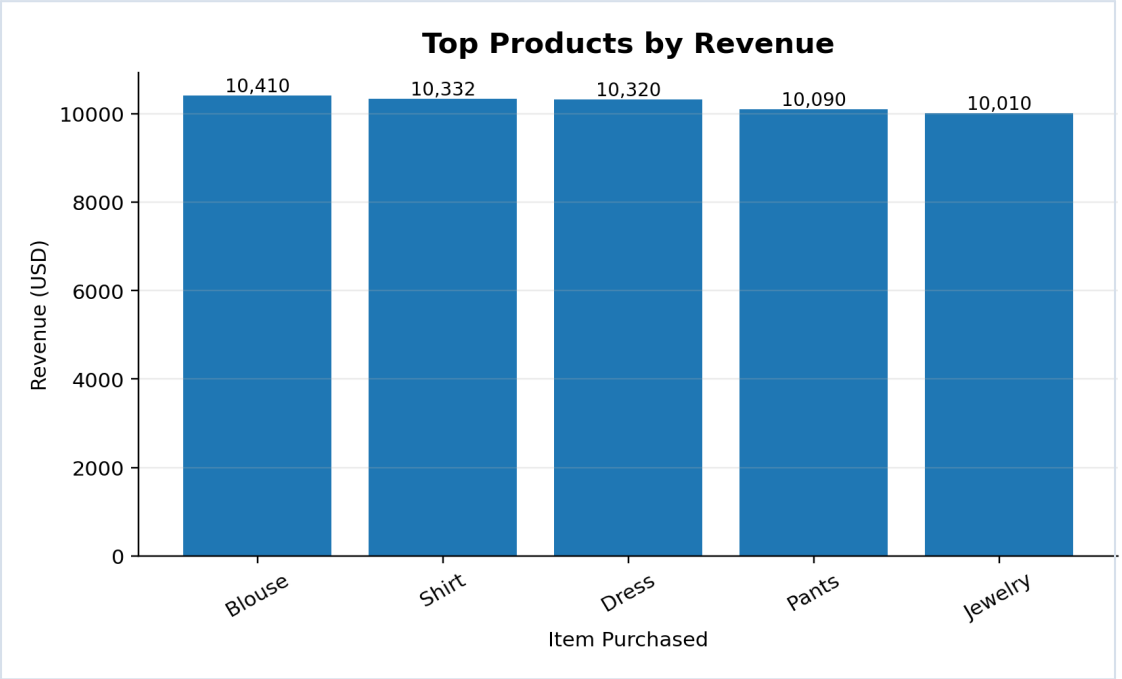


*Figure 3. Top products by revenue. High-volume items are the best candidates for targeted promotions and bundles.*

## 4. Customer, Subscription, and Segment Analysis

Demographic analysis confirms a strong skew toward male revenue contribution and a balanced revenue distribution across age groups, with Young customers slightly ahead. The loyal segment is by far the largest group, which is a positive sign for retention, but it also means the business should invest in repeat-purchase optimization and loyalty value rather than only acquisition.

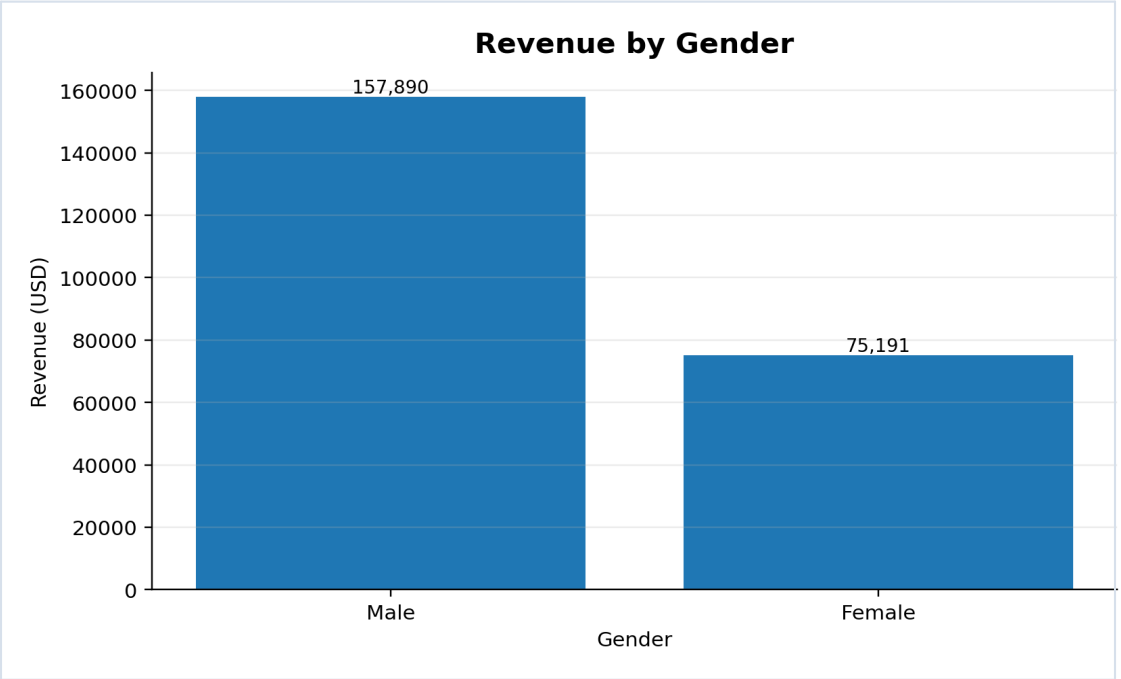


*Figure 4. Revenue by gender. Male customers contribute roughly two-thirds of total revenue.*

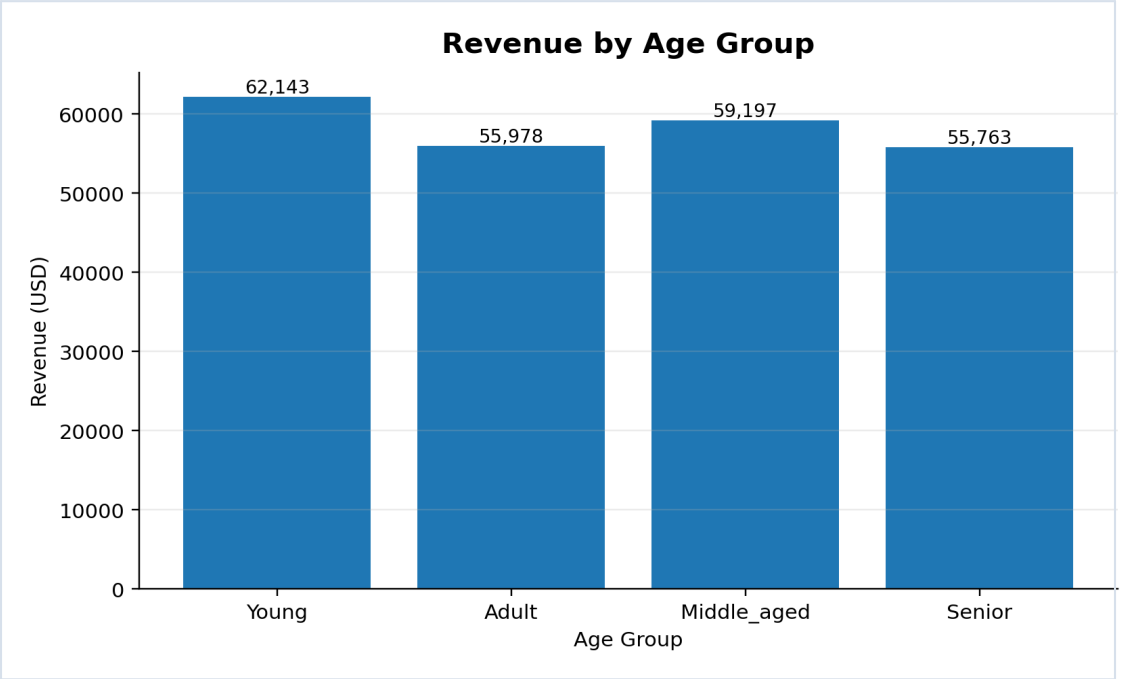


*Figure 5. Revenue by age group. Young and Middle-aged customers are the largest revenue contributors.*

| Gender | Revenue | Share |
|---|---|---|
| Male | $157,890 | 67.7% |
| Female | $75,191 | 32.3% |

| Age group | Revenue | Share |
|---|---|---|
| Young | $62,143 | 26.7% |
| Adult | $55,978 | 24.0% |
| Middle_aged | $59,197 | 25.4% |
| Senior | $55,763 | 23.9% |

| Segment | Customers | Share |
|---|---|---|
| Loyal | 3,116 | 79.9% |
| Returning | 701 | 18.0% |
| New | 83 | 2.1% |

Subscription analysis is notable because the average purchase amount is almost identical for non-subscribers ($59.87) and subscribers ($59.49). Revenue is higher for the non-subscriber group because the base is much larger, not because each customer spends more. That suggests the subscription program may need a sharper benefits story or more exclusive value.

| Subscription | Customers | Avg spend | Revenue |
|---|---|---|---|
| No | 2,847 | $59.87 | $170,436 |
| Yes | 1,053 | $59.49 | $62,645 |

Customer segmentation by previous purchase count shows a strongly loyalty-heavy base: Loyal customers dominate, Returning customers are second, and New customers are small in volume. This is a classic retention-led profile, and it is ideal for lifecycle marketing, loyalty incentives, and personalized re-engagement campaigns.

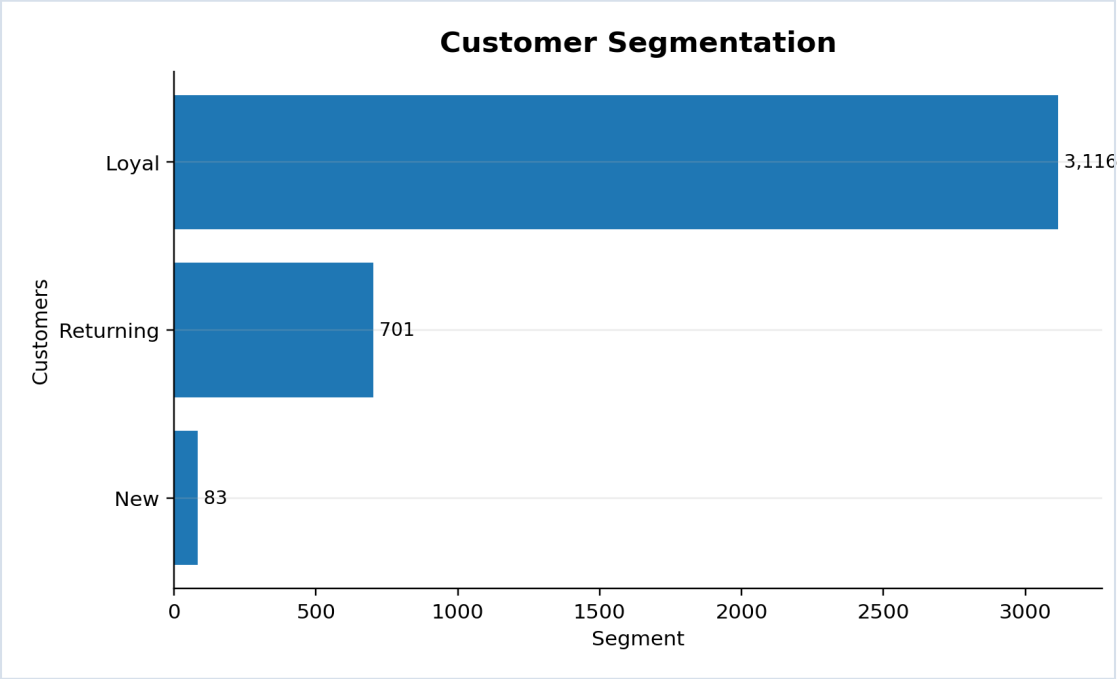


*Figure 6. Customer segmentation. The customer base is heavily weighted toward loyal repeat buyers.*

## 5. Shipping, Discount, and Channel Behavior

Operational and promotional behavior shows that shipping and discounts are useful optimization levers. Revenue differs by shipping method, and discount usage varies by category. These patterns give the business a way to balance conversion, margin, and fulfillment strategy.

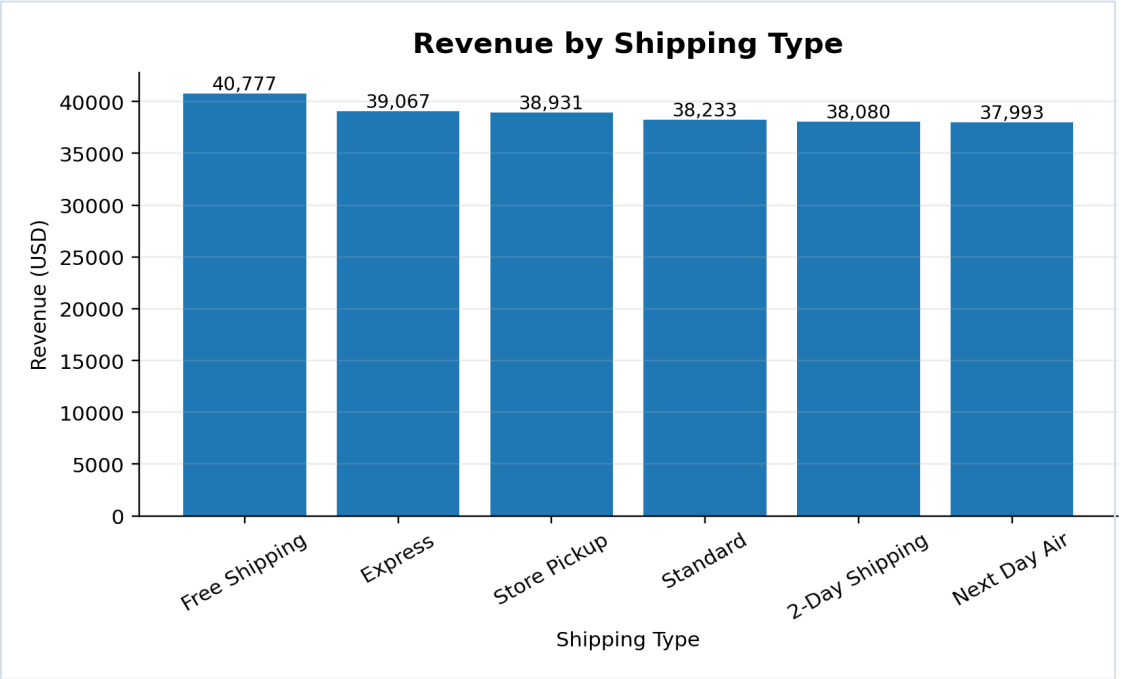


*Figure 7. Revenue by shipping type. Shipping choice influences purchase value and should be reviewed alongside fulfillment cost.*

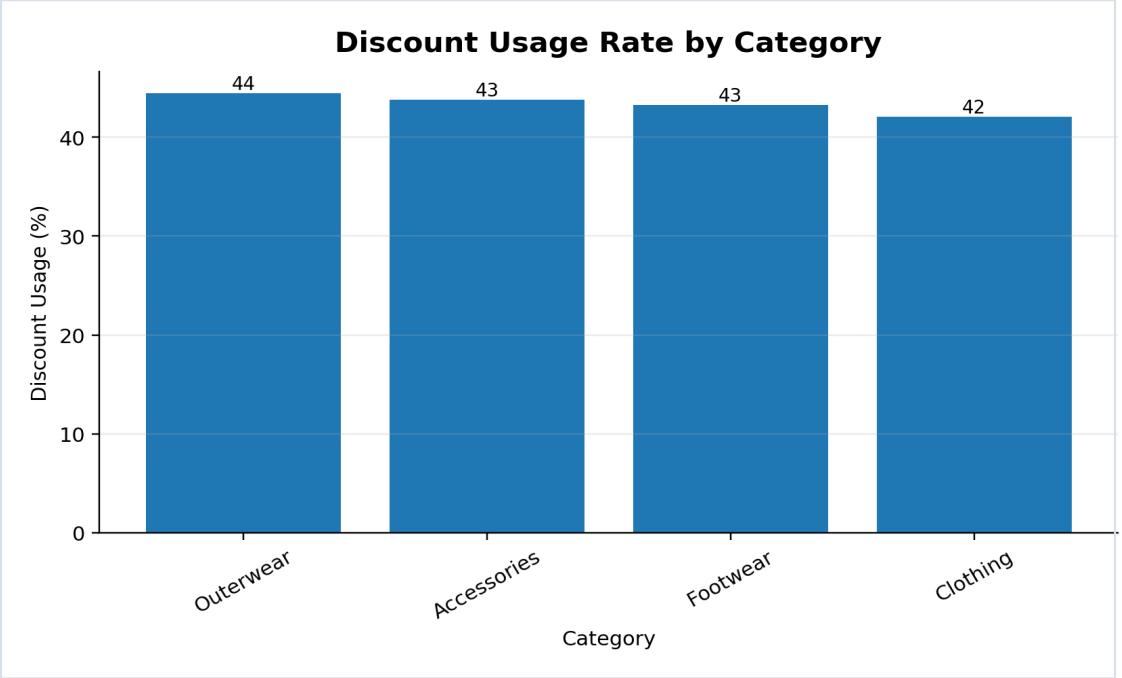


*Figure 8. Discount usage rate by category. Outerwear and Accessories show the highest discount intensity.*

| Shipping type | Orders | Avg spend | Revenue |
|---|---|---|---|
| Free Shipping | 675 | $60.41 | $40,777 |
| Express | 646 | $60.48 | $39,067 |
| Store Pickup | 650 | $59.89 | $38,931 |
| Standard | 654 | $58.46 | $38,233 |
| 2-Day Shipping | 627 | $60.73 | $38,080 |
| Next Day Air | 648 | $58.63 | $37,993 |

| Category | Discount rate | Revenue |
|---|---|---|
| Outerwear | 44.44% | $18,524 |
| Accessories | 43.79% | $74,200 |
| Footwear | 43.24% | $36,093 |
| Clothing | 42.08% | $104,264 |

The strongest practical recommendation from this section is to keep discounts selective. Since some categories already have high discount intensity, broad discounting would likely erode margin without generating a proportional lift in spend. Instead, discount campaigns should be targeted to either high-volume items or categories with lower average spending.

## 6. Dashboard Evidence

The Power BI dashboard was designed to support interactive drilling by subscription status, gender, category, and shipping type. The views below demonstrate that the report is not a static visualization - the same model supports multiple decision scenarios.

Overview dashboard showing KPI cards and category distribution

No subscription filter applied: broad customer base view

Subscribed customers with gender, category, and shipping filters applied

Male + Clothing + Express scenario demonstrating drill-down behavior

*Figure 9. Dashboard gallery. The first view is the full overview; the others show filtered states that confirm the slicers are working correctly and that the insights remain consistent across segments.*

| | customer_segment text | number_of_customers bigint |
|---|---|---|
| 1 | loyal | 3116 |
| 2 | Returning | 701 |
| 3 | New | 83 |

Customer segmentation output: Loyal, Returning, New

| | subscription_status text | total_customers bigint | spend_amount numeric | total_revenue numeric |
|---|---|---|---|---|
| 1 | No | 2847 | 59.87 | 170436.00 |
| 2 | Yes | 1053 | 59.49 | 62645.00 |

Subscription vs non-subscription output

| | gender text | revenue numeric |
|---|---|---|
| 1 | Female | 75191 |
| 2 | Male | 157890 |

Revenue by gender output

| | age_group text | total_revenue numeric |
|---|---|---|
| 1 | Young | 62143 |
| 2 | Middle_aged | 59197 |
| 3 | Adult | 55978 |
| 4 | Senior | 55763 |

Revenue by age group output

*Figure 10. Query output gallery. These screenshots verify the SQL outputs used in the report: segmentation, subscription comparison, revenue by gender, and revenue by age group.*

# 7. SQL Appendix

Below is the query set used to produce the analytical outputs in the report. The structure shows the exact business questions, making the project easy to audit and easy to reuse in GitHub.

## 1. Revenue by gender

```
-- Q1. What is the total revenue generated by male vs female customers?
select gender, sum(purchase_amount) as revenue
from customer
group by gender
order by revenue desc;
```

## 2. Discounted high spenders

```
-- Q2. Which customers used a discount but still spent more than the average purchase amount?
select customer_id, purchase_amount
from customer
where discount_applied = 'Yes'
and purchase_amount >= (
select avg(purchase_amount)
from customer
);
```

## 3. Top rated products

```
-- Q3. Which are the top 5 products with the highest average review rating?
select item_purchased, round(avg(review_rating::numeric), 2) as "Average product Rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5;
```

## 4. Shipping comparison

```
-- Q4. Compare the average purchase amounts between standard and express shipping.
select shipping_type, round(avg(purchase_amount), 2) as average_purchase
from customer
where shipping_type in ('Standard', 'Express')
group by shipping_type;
```

## 5. Subscription comparison

```
-- Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers
and non-subscribers.
select subscription_status,
count(customer_id) as total_customers,
round(avg(purchase_amount), 2) as spend_amount,
round(sum(purchase_amount), 2) as total_revenue
from customer
group by subscription_status;
```

# 7. SQL Appendix (continued)

## 6. Discount sensitivity

```
-- Q6. Which 5 products have the highest percentage of purchases with discount applied?
select item_purchased,
round(100 * sum(case when discount_applied = 'Yes' then 1 else 0 end) / count(*), 2) as discount_rate
from customer
group by item_purchased
order by discount_rate desc
limit 5;
```

## 7. Customer segmentation

```
-- Q7. Segment customers into New, Returning, and Loyal based on total number of previous purchases.
with customer_type as (
select customer_id, previous_purchases,
case
when previous_purchases = 1 then 'New'
when previous_purchases between 2 and 10 then 'Returning'
else 'Loyal'
end as customer_segment
from customer
)
select customer_segment, count(customer_segment) as number_of_customers
from customer_type
group by customer_segment
order by number_of_customers desc;
```

## 8. Top products by category

```
-- Q8. What are the top 3 most purchased products within each category?
with item_counts as (
select category,
item_purchased,
count(customer_id) as total_orders,
row_number() over (partition by category order by count(customer_id) desc) as product_rank
from customer
group by category, item_purchased
)
select product_rank, category, item_purchased, total_orders
from item_counts
where product_rank <= 3;
```

## 9. Repeat buyer behavior

```
-- Q9. Are repeat buyers also likely to subscribe?
select subscription_status,
count(customer_id) as repeat_buyers
from customer
where previous_purchases > 5
group by subscription_status;
```

## 10. Age-group revenue

```
-- Q10. What is the revenue contribution of each age group?
select age_group, sum(purchase_amount) as total_revenue
from customer
group by age_group
order by total_revenue desc;
```

## 8. Business Recommendations

• Reposition subscription around clear value, not just enrollment volume, because average spend is nearly identical for subscribed and non-subscribed customers.

• Prioritize Clothing in merchandising, cross-sell, and campaign allocation because it is the top category by both revenue and transaction count.

• Build retention workflows around loyal customers, who make up the clear majority of the base and are the strongest segment for repeat revenue.

• Target Male, Young, and Middle-aged segments with personalized offers because they are the strongest revenue drivers in the current dataset.

• Use discount policy more selectively in categories with high discount usage to protect margin while preserving conversion.

• Treat shipping method as an operational lever: Standard and Next Day Air underperform on average spend relative to the best-performing methods, so fulfillment strategy should be reviewed alongside customer experience.

The strongest overall message from the project is that the retail business is already retention-heavy and category-led. The next step is not to collect more charts, but to convert these observations into operational actions: tighter subscription offers, category-level campaign prioritization, and loyalty-based retention workflows.

## 9. Conclusion

This project demonstrates an end-to-end analytics workflow from raw retail data to decision-ready insights. The combination of Python, SQL, and Power BI gives the business a clear view of revenue drivers, customer segments, and behavioral patterns. The resulting story is suitable for a GitHub portfolio and a recruiter-facing presentation because it balances technical depth with practical business interpretation.